

Surreal Markov Generators at Every Valuation Scale

Stochastic retracts, prescribed effective chains,
relative stability, and rank-one realization

Research manuscript
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Abstract

A finite matrix of positive surreal transition rates can encode several orders of infinitesimal time scales, including scales that no single real parameter can preserve globally. We study its normalized resolvent rather than assume a global surreal matrix exponential. Directed rooted forests give every entry's valuation and leading coefficient at every scale. The resulting real matrix shadows form a finite nested hierarchy of stochastic projections; each transition between successive projections is exactly the resolvent of an ordinary finite Markov generator on a stochastic retract.

The main converse is a realization theorem: every compatible finite hierarchy of stochastic projections *and prescribed effective Markov generators* occurs at any prescribed strictly ordered Hahn scales in an irreducible surreal rate matrix. An intermediate-scale correction repairs positivity without introducing an additional leading resolvent scale. We also prove an entrywise relative stability bound, uniform over all positive resolvent parameters and without a spectral-gap loss, and show that the entire finite leading-term diagram admits a rank-one monomial realization. Finally, the effective characteristic polynomial recovers the leading surcomplex eigenvalue amplitudes, including tied scales and nonreal eigenvalues. All Hahn calculations use finite algebra in a set-sized workspace; no countable convergence in the full surreal topology is asserted. Throughout, the value group of that workspace is nonzero, set-sized and divisible. Path measures, sample paths and infinite state spaces are not treated.

Status. This manuscript takes the new-theorems alternative, not the claimed solution of a named published open problem. Classical forest identities, Markov-chain aggregation, stochastic-idempotent structure, and rank-one metastability theory are credited explicitly. The arbitrary-rank realization and stability package is presented as a candidate original contribution after a targeted, non-exhaustive comparison. Priority and absence from every repository archive are not certified. The Lean ledger in `docs/FORMALIZATION.md` records checked algebraic clauses of the resolvent identity, projection-flag construction and irreducible completion. The residue calculations and remaining hierarchy theorems are pending. This is an AI-assisted research draft; it has not been refereed. It is built from a single manuscript; Section 1.6 records how that manuscript was placed in the collection and what was changed.

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1 The question, the answer, and its boundaries

A real continuous-time Markov chain has a finite row Laplacian L : its off-diagonal entries are nonpositive and $L\mathbf{1} = 0$. Replacing its rates by positive surreal numbers makes it possible to distinguish transitions of size ω^{-1} , ω^{-2} , and $\omega^{-\omega}$ exactly. This raises a more substantial question than whether familiar matrix identities still hold:

Which finite hierarchies of effective stochastic dynamics can occur when the observation scale moves through an arbitrary ordered group of surreal magnitudes? Can the hierarchy, its rare entries, and its surcomplex spectral amplitudes be recovered without choosing a numerical small parameter?

We answer this for the normalized resolvent

$$R_L(s) = s(sI + L)^{-1}, \quad s > 0. \quad (1.1)$$

For an ordinary real chain, this is the transition matrix observed at an independent exponential time of rate s . In a Hahn field we take (1.1) as a rational-algebraic definition. This choice avoids assuming a globally defined surreal matrix exponential or a surreal-valued probability measure on a path space.

Write $s = ct^\alpha$, where α belongs to the value group and $c > 0$ is real. The central object is

$$K_\alpha(c) = \text{res } R_L(ct^\alpha),$$

where residue means the coefficient of t^0 of a bounded Hahn series. We call the real matrix $K_\alpha(c)$ the *shadow*, or residue matrix, of the resolvent at scale α . Smaller s , hence larger α , corresponds to longer observation. A *plateau* is a range of α on which this matrix is constant in c . A *crossover* is a scale at which nontrivial dependence on c remains. These terms refer only to this precisely defined resolvent hierarchy. Section 2.4 records where these and other words of this report mean something different elsewhere in the collection.

1.1 Main conclusions

There are at most $n - 1$ crossover scales for an irreducible n -state system. Between them lie stochastic idempotents

$$I = P_0, \quad P_1, \dots, P_m = \mathbf{1}\pi_0, \quad P_i P_j = P_j P_i = P_{\max(i,j)}.$$

An idempotent need not be a blockwise averaging matrix: some states can distribute mass among several surviving classes. The correct reduced state space is a *stochastic retract*, a factorization $P = UV$, $VU = I$ with both U and V row stochastic.

At each crossover the shadow is

$$K_\alpha(c) = U c(cI + G_\alpha)^{-1} V,$$

for a real Markov row Laplacian G_α . This matrix is determined by one compressed shadow, $G_\alpha = (VK_\alpha(1)U)^{-1} - I$. The result makes no reversibility, simple-eigenvalue, or unique-minimizing-forest assumption.

The converse is stronger than realizing an eigenvalue list. Prescribe the whole nested family P_j and, on each retract, any real effective generator whose limiting projection is the next P_j . Theorem 7.4

constructs an irreducible Hahn generator with exactly those crossovers at any prescribed ordered scales. Thus the compatibility conditions in the forward theorem are also sufficient.

Two further conclusions sharply delimit what this structure measures. First, relative perturbations of all rates of valuation at least $\delta > 0$ change *every positive resolvent entry* by a relative error of valuation at least δ , uniformly in s . This includes entries whose ordinary residue is zero. Second, every finite leading-term diagram can be reproduced by a real one-parameter monomial model. Higher rank permits exact global distinctions, but no finite diagram of this kind can by itself certify higher rank.

1.2 What is classical, and what is being proposed as new

The directed matrix-forest theorem is classical; our normalization is a change of variable in the identities of Chebotarev–Agaev [3]. Finite stochastic idempotents also have a classical structure theory; a useful modern treatment is Dyer–Greenhill–Ullrich [6]. These are inputs, not new discoveries.

Markov metastability is a substantial existing subject. In particular, Gan–Cameron [4] treat reversible and nonreversible chains, symmetric and nonsymmetric cases, and graph-based critical time scales. Their discussion places eigenvalue-exponent formulas in the Freidlin–Wentzell tradition. Singular-perturbation limits and aggregation are also developed by Yin–Zhang [5] and related work. Merely allowing nonreversible rates, or recovering eigenvalue exponents from optimal forests in rank one, is not claimed as new here. Their path-event and typical-transition-graph hierarchies can contain more scales than the nonzero eigenvalue scales considered in this manuscript. Our bound $n - 1$ does not bound all those other notions of critical time.

Nested self-adjoint projections and prescribed hierarchical spectra have close predecessors in hierarchical Laplacians [7]. Generalized ultrametric inverse M -matrices are another related matrix-theoretic family [8]. Consequently the reversible, partition-based special case of our inverse construction is not proposed as a new general principle. The distinction here is the unrestricted stochastic-retract setting, the simultaneous prescription of each effective generator, and the positivity repair that does not change the leading hierarchy.

Higher-rank Hahn specialization has an established tropical-geometric context [9]. We prove our finite sign-specialization lemma for completeness, not as a new theorem about ordered vector spaces. Its application below preserves all marked-forest leading terms and all effective matrices at once.

Ingredient	Status in this manuscript
Matrix-forest identities	Classical input, extended by polynomial identity.
Stochastic-idempotent splitting	Classical input, with a complete proof.
Forest spectral exponents	Classical rank-one ancestry; arbitrary-rank proof included.
Forward/reverse hierarchy theorem	Candidate contribution: exact compatible-crossover realization over arbitrary rank.
Uniform relative stability	Candidate contribution in its simultaneous, all-parameter and marked-entry formulation.
Finite rank-one realization	Candidate contribution for the full leading resolvent diagram; the separation lemma is classical.

The literature comparison is targeted rather than exhaustive. It did not locate the exact

collection of realization, stability, and simultaneous specialization statements proved here. That is a statement about the search performed, not a proof of historical priority.

1.3 Comparison with the Surreal repository

The manuscript's comparison was pinned to commit

4cf691c7d951e037739d32d9f5c387dcce724f3c

The maintained catalogue contains extensive analytic, dynamical, and spectral material [10]. In particular, its spectral report already treats finite matrices, determinantal scales, perturbations, and regularization as a scale filter. Those subjects are not reclaimed here. Our objects are row-stochastic resolvent shadows, their generally nonorthogonal retracts, and the converse realization of prescribed effective Markov dynamics.

The audit read the main documentation catalogue and the spectral report's inventory, inventoried `docs/new`, and performed an indexed repository search for `Markov`, which returned no matches. The 18 ZIP archives then in `docs/new` were *not* expanded and their full mathematical contents were not searched. Thus the warranted claim is that the package was not found in the inspected material, not an exhaustive assertion about every binary archive. The accompanying `SOURCE_AUDIT.md` records this boundary.

Since the pin. Each of the 18 archives that `docs/new` held at the pin has since been unpacked, placed in the collection and removed from `docs/new`, and so has this manuscript. At commit `e9a9650`, which placed this report, a full-text search of every LaTeX source in the collection finds the words `Markov` and `metastab` only in this report. Apart from bibliography titles, the word `stochastic` occurs in exactly one other report: a proof remark in the spectral report [12], following its Frobenius eigenvalue bound (`thm:HW`), says that the proof does not use a decomposition of a doubly stochastic matrix into permutations. That remark is about a proof technique, not about Markov dynamics. These searches, like the original one, locate words, not results stated under other names.

1.4 What this report does not claim

Each limitation below is also stated where it applies. They are collected here so that none of them is lost when the results are quoted elsewhere.

- (N1) *No global surreal exponential and no path measure.* The object of study is the normalized resolvent, defined as a rational-algebraic expression. No globally defined surreal matrix exponential, surreal heat semigroup, surreal-valued probability measure on a path space, surreal sample path, or stochastic integral is constructed or assumed. The real semigroup $S_\alpha(u)$ of Corollary 6.3 is not claimed to be the residue of a global $\exp(-ut^{-\alpha}L)$ (Remark 6.4). Constructing such objects would require further choices of convergence and summability structure (Section 12.4).
- (N2) *No convergence in the full surreal topology.* Every proof is finite algebra in one set-sized Hahn workspace. Residue is a ring homomorphism, not a sequential limit in the fine topology of **No**. Limits in the real variables c and u are ordinary limits of finite real matrices, and no assertion that t^n tends to zero in an arbitrary-rank Hahn topology is used (Remark 2.1). A

bound $O_v(\rho)$ is a valuation assertion, not a bound with an unspecified real constant, and the remainder certificate of Proposition 4.2 uses neither a metric on the whole surreal class nor a convergence rate in an ordinary norm.

- (N3) *Finite state spaces only.* No result covers countably infinite state spaces, time-inhomogeneous rates, uniform estimates as $n \rightarrow \infty$, or a full hierarchy of large-deviation transition events.
- (N4) *Divisibility is assumed throughout.* Section 2.3 lists where it is used. No statement is claimed for a nondivisible value group.
- (N5) *One notion of critical time.* Plateaux and crossovers refer only to the resolvent hierarchy. The bound $n - 1$ bounds these crossover scales, which are the distinct valuations of nonzero eigenvalues. It does not bound the path-event or typical-transition-graph hierarchies of the metastability literature, which can contain more scales.
- (N6) *Rank-one realization preserves finite leading data only.* Theorem 9.2 preserves the finite marked-forest leading diagram. It does not preserve complete Hahn series, all possible inequalities, or the ordered group, and it is not a global order embedding of Γ (Remark 9.3). Higher rank remains meaningful for unbounded families of scale comparisons, for complete series, and for data whose size is not fixed in advance; the finite theorem is not to be extrapolated to those settings.
- (N7) *The stability hypotheses matter.* Theorem 8.1 needs positive rates on one fixed graph. A new edge is not a relative perturbation of a zero edge, and the theorem is not a condition-number bound for arbitrary matrices (Remark 8.4).
- (N8) *The leading spectrum need not be real.* No assertion is made that a nonreversible spectrum is real: leading amplitudes can form nonreal conjugate pairs (Example 11.2). Diagonalizability at nonzero eigenvalues is neither assumed nor proved.
- (N9) *Computation.* Forest enumeration is finite, not efficient. It needs an exact comparison contract for the input valuations and coefficients, and a finiteness theorem is not a uniform algorithm on every representation of a surreal number (Section 12.1). The finite checks of Section 12.2 do not establish the general theorems.
- (N10) *Credited, not claimed.* The directed matrix-forest theorem (Chebotarev–Agaev), the structure of stochastic idempotents (Dyer–Greenhill–Ullrich), metastability with nonreversible rates and optimal-forest eigenvalue exponents in rank one (Gan–Cameron, in the Freidlin–Wentzell tradition), singular-perturbation aggregation (Yin–Zhang), nested self-adjoint projections with prescribed hierarchical spectra (Bendikov–Krupski), generalized ultrametric inverse M -matrices (Nabben–Varga), and higher-rank Hahn specialization (Joswig–Smith) are inputs or context. The reversible partition case of the inverse construction and the finite rational sign-separation lemma are not proposed as new.
- (N11) *Priority and verification.* The realization and stability package is a candidate original contribution after a targeted, non-exhaustive literature comparison. Priority is not certified, and neither is absence from every archive of the repository at the pin (Section 1.3). The opening status paragraph identifies the algebraic clauses checked in Lean; it does not claim formal verification of the full hierarchy. The report has not been refereed. The three further questions of Section 12.4 are not solved.

The delivered audit file `SOURCE_AUDIT.md` states the audit boundaries in more detail.

1.5 Relation to other reports of the collection

No proof in this report depends on another report, and no other report depends on this one. The links below are comparisons.

Matrix scaling. The report `docs/surreal/matrix-scaling-at-surreal-scales/` uses the same no-cancellation mechanism for undirected spanning-tree sums (`scale:thm:tree`). Its deletion gap κ_e gives relative gains $\delta + \kappa_e$ for a fixed-margin normalization (`scale:thm:sharp`), unlike the gain-free Theorem 8.1 here; the two concern different maps.

Newton’s rule in the polynomial-algebra report. The polynomial-algebra report [11] works, like this one, over $K = \mathbb{C}((t^\Gamma))$ with Γ set-sized and divisible. It proves Newton’s root-valuation rule at arbitrary rank (`polynomial:thm:newton`), together with the initial-polynomial root count and the profile formula $\phi_P(\rho) = v(a_n) + \sum_i \min\{\rho, v(\alpha_i)\}$ (`polynomial:thm:initialroots` and `polynomial:eq:profileproduct`). Apply these at center zero to the monic forest polynomial $D(s) = \sum_k \sigma_k s^{d-k} = \prod_r (s + \lambda_r)$ of (3.3). Its coefficient of s^{d-k} has valuation h_k , so its profile at α is exactly m_α of (4.1), and they give the three assertions of Lemma 10.1: the valuation identity (10.1), the initial factorization (10.2), and the identification of the critical scales with the valuations of nonzero eigenvalues. The direct proof given here is kept. Two features are specific to rate matrices. First, the coefficient valuations h_k are read from forests with no possible cancellation, because every forest weight is positive. Second, Theorem 10.3 proves more than Newton’s rule gives for a general polynomial: every point (k, h_k) lies on the lower boundary, since the increments are nondecreasing, and at a critical scale every index from k_- to k_+ is active. For a general polynomial some coefficient points may lie strictly above the lower boundary. Here the positivity of the forest coefficients of the effective generator rules that out.

The positive Cauchy–Binet identity of the spectral report. The spectral report’s positive Cauchy–Binet identity (`thm:gram`, a label without report prefix in that report) writes the coefficients of $\det(I + sA^*A)$ as sums of squared moduli of minors. Because these are positive, no leading cancellation occurs, and the coefficient valuations are twice the minimal minor valuations. Equation (3.10) uses the same mechanism for a different positive expansion: by Proposition 3.1, the coefficients σ_k of $\det(sI + L)/s$ are sums of positive forest weights, so $v(\sigma_k) = h_k$ and $\text{lc}(\sigma_k) = b_k > 0$. The matrix classes differ. A Gram matrix is Hermitian and positive semidefinite, and its determinantal scales are singular-value scales. A row Laplacian is in general not normal, and its eigenvalues can be nonreal (Example 11.2). Neither result implies the other. The same report fixes the word *spectrum* to mean the algebraic eigenvalue list of a finite matrix over K together with the valuations (`spec:sec:spectrumword`). Section 10 uses the word in that sense, for the row Laplacian L .

Measures and probability. The collection’s report on measures and probability is `docs/surreal/hahn-valued-measures-and-probability/` [13]. This report does not use it, and it does not use this report. The path measures, sample paths and infinite state spaces excluded in (N1) and (N3) are not supplied here by an appeal to that report. Nothing here asserts or denies that its constructions apply to the rate matrices studied here.

1.6 Provenance of this report

This report is built from one manuscript, *Surreal Markov Generators at Every Valuation Scale*, dated 22 September 2026. It arrived as the archive `surreal_markov_hierarchies`, with LaTeX source `surreal_markov_hierarchies.tex`, the audit file `SOURCE_AUDIT.md`, the verification program `verify.py` with its recorded output, a build record, a requirements file and a build script. It came with the second of two deliveries, repository commit `15d4bab`, and was placed at commit `e9a9650`. Its own repository comparison is pinned to `4cf691c7d951e037739d32d9f5c387dcce724f3c` (Section 1.3).

There is no second source, so there was no merge and no choice between sources. Nothing was selected out of a larger body of work. Every theorem, proof, example and disclaimer of the manuscript is kept. The report is a standalone member of the surreal family because its rates, plateau projections and effective generators are real Hahn objects in $F = \mathbb{R}((t^\Gamma))$. The surcomplex eigenvalue amplitudes of Section 10 are a consequence of that real theory.

Placement changed the following and nothing else.

- (a) The source was renamed `article.tex`. The audit file, the program, its recorded output and the build record are kept byte-identical. The build script was moved to `code/` and the requirements file to `data/`.
- (b) Every label now carries the prefix `markov:`. All 100 labels of the manuscript were renamed in the same way, and none was dropped.
- (c) Sections 1.4, 1.5, 1.6, 2.3 and 2.4 were added, and so were Remarks 3.3 and 10.5, which point to other reports. They were placed so that every theorem, equation and section number of the manuscript is unchanged. The abstract and the status paragraph now state the divisibility hypothesis, the exclusion of path measures, and the draft status. Sections 9 and 10 now say where they use divisibility. One sentence defining the shadow was added to Section 1, because the manuscript used the word without defining it. One sentence in Section 12.4 points to the measure report, and one paragraph in Section 12.2 records how to rerun the checks.
- (d) Repository statements that had gone stale were corrected, with the pin kept as provenance (Section 1.3).
- (e) In Section 12.2, the description of what the program compares was made exact on one point (multiplicities).

Placement left the mathematics unchanged. Subsequent proof corrections are recorded in the maintained README. The source archive, its PDF and its README are not shipped.

2 Hahn workspaces and positive rate matrices

2.1 A set-sized setting inside the surreals

Fix a nonzero, set-sized, divisible ordered abelian group Γ . Put

$$F = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)).$$

A Hahn series has well-ordered support in Γ , and multiplication is the usual well-defined convolution. For $x \neq 0$, set

$$v(x) = \min \text{supp } x, \quad \text{lc}(x) = [t^{v(x)}]x, \quad v(0) = +\infty.$$

The order on F is determined by the sign of its leading coefficient. Its valuation ring and maximal ideal are

$$\mathcal{O} = \{x : v(x) \geq 0\}, \quad \mathfrak{m} = \{x : v(x) > 0\}.$$

The residue map $\text{res} : \mathcal{O} \rightarrow \mathbb{R}$ takes the t^0 coefficient. For bounded complex series its target is $\mathbb{C}$. A positive element of F is bounded by a real constant exactly when its valuation is nonnegative. All nonzero real numbers have valuation zero.

The standard Hahn-field facts used below are that F is real closed and K is algebraically closed when Γ is divisible. The identification $K = F[i]$ is coefficientwise. Conway normal form [1, 2] supplies the connection to surreal numbers: for an ordered subgroup $\Gamma \subseteq \mathbf{No}$, the map

$$\sum_{\gamma} a_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} a_{\gamma} \omega^{-\gamma} \quad (2.1)$$

is the corresponding normal-form embedding. The minus sign reverses the increasing Hahn support into a decreasing Conway support.

Any finite collection of surreal or surcomplex coefficients is contained in one such workspace: take the union of their normal-form supports (using real and imaginary parts for surcomplex inputs), change signs, adjoin 1, and take the rational span. Each support is a set, though it need not be finite. The resulting span is a nonzero set-sized divisible ordered subgroup of $\mathbf{No}$; adjoining 1 also covers purely real inputs, whose support exponents alone could give the zero group. All subsequent operations take place in its Hahn field or, where stated, a larger set-sized one. No set of all surreal numbers is formed.

Remark 2.1 (No hidden convergence assertion). Every determinant, forest sum, and matrix inverse in this paper is finite algebra. Residue is a ring homomorphism, not a sequential limit in the fine topology of $\mathbf{No}$. Limits in a real variable c or u later in the paper are ordinary limits of finite real matrices. No assertion that t^n tends to zero in an arbitrary-rank Hahn topology is used.

2.2 Rates, leading data, and conventions

Let E be a strongly connected directed graph on $\{1, \dots, n\}$, with $n \geq 2$, without loops. For every allowed edge $e = (i, j)$ choose $q_e \in F_{>0}$. Set $q_{ij} = 0$ on missing edges. The *row Laplacian* is

$$L_{ij} = -q_{ij} \quad (i \neq j), \quad L_{ii} = \sum_{j \neq i} q_{ij}. \quad (2.2)$$

The Markov generator in the usual sign convention is $-L$. All matrices act on column observables; distributions are row vectors. In particular $L\mathbf{1} = 0$.

Each existing rate has a unique leading representation

$$q_e = a_e t^{\gamma_e} (1 + \varepsilon_e), \quad a_e > 0 \text{ real}, \quad \gamma_e \in \Gamma, \quad v(\varepsilon_e) > 0, \quad (2.3)$$

allowing $\varepsilon_e = 0$. The coefficients beyond a_e need not be positive. The leading data are the finite family $(a_e, \gamma_e)_{e \in E}$.

For $\rho \in \Gamma \cup \{+\infty\}$ write $x = O_v(\rho)$ to mean $v(x) \geq \rho$; for a matrix this means the bound entrywise. In particular, $x = O_v(+\infty)$ means $x = 0$, including the zero matrix in the matrix case. This is a valuation assertion, not a bound with an unspecified real constant. Products of finitely many factors $1 + O_v(\rho)$, for $\rho > 0$, again have the form $1 + O_v(\rho)$.

2.3 Where divisibility of the value group is used

Divisibility of Γ is a standing hypothesis of every section. The text uses it explicitly in the following places.

- (D1) The candidate critical scales (5.1) are quotients $(h_\ell - h_k)/(\ell - k)$. They lie in Γ because Γ is divisible. The crossing points of the forest arrangement in Section 9 are quotients of the same kind.
- (D2) Step 2 of the proof of Theorem 7.4 chooses scales β_j strictly between α_{j-1} and α_j , for example their midpoint.
- (D3) Section 9 uses the rational span $W = \text{span}_{\mathbb{Q}}\{\gamma_e\}$ as a subgroup of Γ .
- (D4) Section 10 places the eigenvalues of L in K , which is algebraically closed because Γ is divisible. Lemma 10.1 and Theorems 10.2 and 10.3 rely on this.
- (D5) The surreal transfer above takes the rational span of the set of normal-form exponents of finitely many coefficients, after changing signs and adjoining 1. This set can be infinite, unlike the finite family of leading edge exponents used to define W . Its rational span is still a set-sized divisible subgroup of **No**.

The report claims none of its statements for a nondivisible value group. It does not determine which of them would survive without divisibility.

2.4 Words and symbols used differently elsewhere in the collection

Several words of this report mean other things in other reports of the collection. Each has one meaning here.

- *Shadow* means the real residue matrix $K_\alpha(c) = \text{res } R_L(ct^\alpha)$ of the normalized resolvent, and its analogues for other matrices. It has nothing to do with the “shadow topology”, which the collection’s notation guide [15] lists as a local name for the standard-part topology. This report puts no topology on F . In prose about several reports, “residue matrix” should be used instead.
- *Residue* res is the t^0 coefficient on $\mathcal{O}$, applied entrywise to matrices. It is the scalar standard part st of the notation guide, which lists res as an alias on that domain. It is not a contour or local residue.
- *Flag* means the nested chain of stochastic idempotents $P_0 = I, P_1, \dots, P_m = \mathbf{1}\pi_0$ with $P_i P_j = P_j P_i = P_{\max(i,j)}$. These are the plateaux of the projection hierarchy, Theorem 5.2, whose label keeps the manuscript’s word, and the input of Theorem 7.1. It has nothing to do with the chain of radicals of a lexicographic bilinear form that the multiscale theta-series material in the Hahn–Tate report [14] also calls a flag. The report prefixes keep the two labels apart.
- *Profile* always means the forest profile $(h_k)_{0 \leq k \leq d}$ of (3.7). It is the Newton profile of the forest polynomial D (Section 1.5). Other reports use the word for other finite data.

- The symbols h_k , b_k , B_k , $D(s)$, $D_\alpha(c)$, $N(s)$ and $N_\alpha(c)$ are local. In particular h_k is the minimal forest weight, and D and D_α are forest denominators. Other reports use these letters for unrelated quantities. In prose about several reports, the words “minimal forest weight” and “forest denominator” should be used instead of the symbols.
- *Resolvent* means the normalized $R_L(s) = s(sI + L)^{-1}$ with $s > 0$ in F . Elsewhere in the collection, for instance in the two spectral reports, a resolvent is the unnormalized inverse $(\zeta I - A)^{-1}$ or $(A - \zeta I)^{-1}$. Here $R_L(s) = s(L - \zeta I)^{-1}$ at $\zeta = -s$.
- *Eigenvalues* are those of the row Laplacian L , not of the conventional generator $-L$. Nonzero eigenvalues therefore have positive real parts.
- Theorem 8.1 is a relative stability bound for resolvent entries. It shares only its local label name with the root-stability theorem of the polynomial-algebra report (`polynomial:thm:stability`).

3 The finite forest algebra

An *in-forest* is a spanning directed forest in which every nonroot vertex has exactly one outgoing edge and following outgoing edges leads to a root. Thus its edges point toward the roots. Let $\mathcal{F}_k$ be the finite set of such forests with k edges, and put $d = n - 1$. For $f \in \mathcal{F}_k$, let $r_f(i)$ be the root reached from vertex i and let

$$q(f) = \prod_{e \in f} q_e.$$

The empty forest has weight one. Strong connectivity guarantees that $\mathcal{F}_d \neq \emptyset$, and deleting edges gives $\mathcal{F}_k \neq \emptyset$ for every $0 \leq k \leq d$.

Define the scalar and matrix forest coefficients

$$\sigma_k = \sum_{f \in \mathcal{F}_k} q(f), \quad F_k(i, j) = \sum_{\substack{f \in \mathcal{F}_k \\ r_f(i)=j}} q(f). \quad (3.1)$$

Every row of F_k sums to σ_k , and every forest has $n - k$ roots. Hence

$$F_k \mathbf{1} = \sigma_k \mathbf{1}, \quad \text{tr } F_k = (n - k) \sigma_k, \quad F_0 = I, \quad \sigma_0 = 1. \quad (3.2)$$

Proposition 3.1 (Classical matrix-forest identity). *For the convention (2.2),*

$$\det(sI + L) = sD(s), \quad D(s) = \sum_{k=0}^d \sigma_k s^{d-k}, \quad (3.3)$$

$$\text{adj}(sI + L) = N(s), \quad N(s) = \sum_{k=0}^d F_k s^{d-k}. \quad (3.4)$$

Consequently, for every positive $s \in F$,

$$R_L(s) = \frac{N(s)}{D(s)}. \quad (3.5)$$

It is an entrywise positive row-stochastic matrix. In particular all its entries belong to $\mathcal{O}$ and admit real residues.

Proof. The determinant and adjugate identities are the directed matrix-forest theorem [3, Propositions 2 and 3], with the same row-Laplacian and in-forest orientation. They are polynomial identities with integer coefficients in the edge weights and s . Their differences vanish on the open positive orthant of real weights and positive s , so these polynomials are identically zero. Thus the identities hold in any characteristic-zero coefficient field; alternatively one may use the usual cycle-cancellation proof with formal weights. There is no specialization or continuity assumption on Hahn series in this step.

For $s > 0$, all terms in $D(s)$ are positive, so $sI + L$ is invertible and cancellation of s gives (3.5). Equation (3.2) gives row sum one. For every i, j , a spanning tree rooted at j contributes to $N(s)_{ij}$, so that entry is strictly positive. It is at most one because the other entries in its row are nonnegative. This proves boundedness. $\square$

Remark 3.2. We use the same formulas for a reducible row Laplacian when needed. Some maximal forest coefficients may then be zero, but the determinant has a positive s^n term and the resolvent is still nonnegative and row stochastic for $s > 0$. Irreducibility is needed for strict positivity of every entry, not for the rational identity itself.

3.1 Leading forest profiles

For a forest define its valuation cost and amplitude by

$$w(f) = \sum_{e \in f} \gamma_e, \quad A(f) = \prod_{e \in f} a_e. \quad (3.6)$$

For each edge count define

$$h_k = \min_{f \in \mathcal{F}_k} w(f), \quad (3.7)$$

$$b_k = \sum_{\substack{f \in \mathcal{F}_k \\ w(f) = h_k}} A(f), \quad (3.8)$$

$$B_k(i, j) = \sum_{\substack{f \in \mathcal{F}_k \\ w(f) = h_k \\ r_f(i) = j}} A(f). \quad (3.9)$$

There is no cancellation among leading terms because all amplitudes are positive. Thus

$$v(\sigma_k) = h_k, \quad \text{lc}(\sigma_k) = b_k > 0, \quad B_k \mathbf{1} = b_k \mathbf{1}, \quad \text{tr } B_k = (n - k)b_k. \quad (3.10)$$

Although $B_k = \text{res}(t^{-h_k} F_k)$, its zero entries may conceal higher-valuation marked forests. They will be recovered separately rather than discarded.

Remark 3.3 (The same mechanism in the spectral report). That positivity forbids leading cancellation is also the mechanism of the positive Cauchy–Binet identity (**thm:gram**) of the collection’s spectral report [12]. There it is applied to squared minors of a Gram matrix, here to forest weights of a row Laplacian. Section 1.5 compares the two.

4 Every scale and every marked entry

For $\alpha \in \Gamma$ define

$$m_\alpha = \min_{0 \leq k \leq d} \{h_k + (d - k)\alpha\}, \quad (4.1)$$

$$I_\alpha = \{k : h_k + (d - k)\alpha = m_\alpha\}, \quad (4.2)$$

$$D_\alpha(c) = \sum_{k \in I_\alpha} b_k c^{d-k}, \quad N_\alpha(c) = \sum_{k \in I_\alpha} B_k c^{d-k}. \quad (4.3)$$

The expression m_α is also the minimum of $w(f) + (d - |f|)\alpha$ over all forests. Its definition uses only ordered-group addition and comparison; drawing a Newton polygon in a real plane is neither necessary nor, at arbitrary rank, a literal interpretation.

Theorem 4.1 (The complete leading-forest formula). *For every $\alpha \in \Gamma$ and real $c > 0$,*

$$K_\alpha(c) := \text{res } R_L(ct^\alpha) = \frac{N_\alpha(c)}{D_\alpha(c)}. \quad (4.4)$$

This is a real, nonnegative, row-stochastic rational matrix function of c . More generally, every positive $s \in F$ has a unique form $s = ct^\alpha(1 + \eta)$ with $c > 0$ real and $v(\eta) > 0$, and its resolvent has the same residue $K_\alpha(c)$.

For a fixed entry define

$$m_{ij,\alpha} = \min_{\substack{f \in \bigcup_k \mathcal{F}_k \\ r_f(i)=j}} \{w(f) + (d - |f|)\alpha\}, \quad (4.5)$$

$$A_{ij,\alpha}(c) = \sum_{\substack{r_f(i)=j \\ w(f)+(d-|f|)\alpha=m_{ij,\alpha}}} A(f)c^{d-|f|}. \quad (4.6)$$

Then

$$v(R_L(ct^\alpha)_{ij}) = m_{ij,\alpha} - m_\alpha, \quad (4.7)$$

$$\text{res}\left(t^{-(m_{ij,\alpha}-m_\alpha)} R_L(ct^\alpha)_{ij}\right) = \frac{A_{ij,\alpha}(c)}{D_\alpha(c)} > 0. \quad (4.8)$$

In particular the data (a_e, γ_e) determine the leading term of every entry, including entries whose residue is zero.

Proof. A denominator forest term $q(f)s^{d-|f|}$ has valuation $w(f) + (d - |f|)\alpha$ and leading coefficient $A(f)c^{d-|f|}$. A finite sum of positive Hahn elements has valuation the minimum of their valuations, and its leading coefficient is the sum of the positive leading coefficients at that minimum. Therefore

$$\text{res}(t^{-m_\alpha} D(ct^\alpha)) = D_\alpha(c) > 0, \quad \text{res}(t^{-m_\alpha} N(ct^\alpha)) = N_\alpha(c).$$

The normalized denominator is a unit of $\mathcal{O}$. Taking its inverse and then residue proves (4.4). The row-stochastic assertion follows either from Proposition 3.1 or from $N_\alpha(c)\mathbf{1} = D_\alpha(c)\mathbf{1}$.

The numerator for a marked entry is another nonempty finite positive forest sum. Normalizing it at its own minimum gives (4.7)–(4.8). Finally, $(1 + \eta)^{d-|f|}$ has residue one for every forest. Replacing ct^α by $ct^\alpha(1 + \eta)$ therefore changes none of the leading computations. $\square$

Proposition 4.2 (A valuation remainder certificate). *Let*

$$\delta = \min_{e \in E} v(\varepsilon_e),$$

where $\delta = +\infty$ if all corrections vanish. At a fixed α , let

$$\Delta_\alpha = \min\{w(f) + (d - |f|)\alpha - m_\alpha : w(f) + (d - |f|)\alpha > m_\alpha\},$$

with minimum $+\infty$ for the empty set. Then, for every real $c > 0$,

$$R_L(ct^\alpha) - K_\alpha(c) = O_v(\min(\delta, \Delta_\alpha)). \quad (4.9)$$

The valuation lower bound is independent of c . If both quantities are infinite, the identity is exact.

Proof. Every forest weight is $A(f)t^{w(f)}(1 + O_v(\delta))$. After division by t^{m_α} , nonminimal leading forest terms have valuation at least Δ_α and corrections to minimal terms have valuation at least δ . Thus the normalized numerator and denominator equal their real initial polynomials plus errors of valuation at least their minimum. The real denominator $D_\alpha(c)$ is nonzero, so inversion of this unit preserves the error bound. All powers of the nonzero real c have valuation zero. $\square$

This certificate distinguishes two sources of uncertainty: subleading terms inside an edge rate, and nonminimal forests. Neither a metric on the whole surreal class nor a convergence rate in an ordinary norm appears in it.

5 The finite flag of stochastic projections

Call α *critical* if I_α contains at least two indices. There are only finitely many critical values: every possible one belongs to

$$\left\{ \frac{h_\ell - h_k}{\ell - k} : 0 \leq k < \ell \leq d \right\}. \quad (5.1)$$

Division by a positive integer is available because Γ is divisible. Away from these values the active index is unique and $K_\alpha(c) = B_k/b_k$ is independent of c .

Lemma 5.1 (Resolvent compatibility). *For positive $s \neq u$,*

$$R_L(s)R_L(u) = \frac{uR_L(s) - sR_L(u)}{u - s}. \quad (5.2)$$

Consequently

$$K_\alpha(c)K_\alpha(e) = \frac{eK_\alpha(c) - cK_\alpha(e)}{e - c} \quad (c, e > 0, c \neq e), \quad (5.3)$$

$$K_\alpha(c)K_\beta(e) = K_\beta(e)K_\alpha(c) = K_\beta(e) \quad (\alpha < \beta). \quad (5.4)$$

Proof. The two resolvents commute since they are rational functions of L . Subtracting $(sI + L)^{-1}$ and $(uI + L)^{-1}$ proves (5.2). Substituting $s = ct^\alpha$, $u = et^\alpha$ and taking residues gives (5.3). If $\alpha < \beta$, divide numerator and denominator on the right of (5.2) by s . The ratio u/s has positive valuation; all resolvent entries are bounded. The residue of $((u/s)R_L(s) - R_L(u))/((u/s) - 1)$ is therefore $K_\beta(e)$. $\square$

Theorem 5.2 (Projection hierarchy). *There are critical scales $\alpha_1 < \dots < \alpha_m$, with $1 \leq m \leq n - 1$, and distinct stochastic idempotents*

$$P_0 = I, \quad P_1, \dots, P_m = \mathbf{1}\pi_0, \quad P_i P_j = P_j P_i = P_{\max(i,j)}, \quad (5.5)$$

where π_0 is a real probability row vector, possibly with zero entries. On the open interval between α_j and α_{j+1} the shadow is P_j ; the initial and final intervals have shadows I and $\mathbf{1}\pi_0$. The ranks decrease strictly.

At a critical scale α , write

$$k_- = \min I_\alpha, \quad k_+ = \max I_\alpha, \quad p = n - k_-, \quad q = n - k_+.$$

The preceding and following plateaux satisfy

$$P = \lim_{c \rightarrow \infty} K_\alpha(c) = B_{k_-}/b_{k_-}, \quad (5.6)$$

$$Q = \lim_{c \downarrow 0} K_\alpha(c) = B_{k_+}/b_{k_+}, \quad (5.7)$$

with $\text{rank } P = p$, $\text{rank } Q = q$, and

$$PK_\alpha(c) = K_\alpha(c)P = K_\alpha(c), \quad K_\alpha(c)Q = QK_\alpha(c) = Q. \quad (5.8)$$

Both displayed limits are limits of real rational matrices.

Proof. At a noncritical scale $K_\alpha(c) = P$ for every c . Taking two different real values in (5.3) gives $P^2 = P$. If k is the active index, (3.10) gives

$$\text{rank } P = \text{tr } P = n - k,$$

since the eigenvalues of an idempotent are zero and one. Cross-scale compatibility proves all products in (5.5).

As α increases, a lower envelope of lines with slopes $d - k$ can only move to larger k . More explicitly, if k is active at α and ℓ is active at $\beta > \alpha$, adding their two minimizing inequalities gives $(\ell - k)(\beta - \alpha) \geq 0$. Thus $\ell \geq k$. Choosing α below all the finitely many crossing values makes $k = 0$ the unique minimizer; choosing it above all of them makes $k = d$ the unique minimizer. These choices exist in any nonzero ordered group. The first matrix is I . The last has rank one, is stochastic, and so has identical probability rows.

At a critical value, the highest power of c in both initial polynomials comes from k_- and the lowest comes from k_+ . This proves the endpoint limits. Immediately before and after that value, the corresponding unique minimizers are k_- and k_+ . Since $k_- < k_+$, the rank drops by a positive integer at each critical value. The total rank loss is $n - 1$, proving $m \leq n - 1$. Finally, apply (5.4) using any scale in the adjacent plateau intervals to obtain (5.8). $\square$

Remark 5.3 (What the last row vector means). The exact stationary distribution is given by the tree formula $\pi_j = F_d(i, j)/\sigma_d$, independently of i . Hence $\pi_0 = \text{res } \pi$. Exact irreducibility makes every π_j positive, but does not force every $\text{res } \pi_j$ to be positive. Infinitesimally occupied states explain the zero columns and transient mixing that can occur in intermediate projections.

6 Every crossover is a real effective Markov chain

6.1 Stochastic retracts, including transient states

Lemma 6.1 (Classical splitting of a stochastic idempotent). *If P is an $n \times n$ real row-stochastic idempotent of rank p , there exist nonnegative row-stochastic matrices*

$$U \in \mathbb{R}^{n \times p}, \quad V \in \mathbb{R}^{p \times n}, \quad P = UV, \quad VU = I_p.$$

A canonical choice, up to ordering, is indexed by the closed communicating classes of P : the rows of V are their stationary probability vectors, extended by zero, and U_{ia} is the mass that row i assigns to class a .

Proof. Here is a direct proof of the standard structure result [6]. Decompose the directed graph of positive entries of P into its closed communicating classes C_a and its remaining states T . With closed classes first, its matrix has the form

$$P = \begin{pmatrix} C & 0 \\ B & T_0 \end{pmatrix},$$

where C is block diagonal with irreducible stochastic blocks. Every transient state has a finite positive-probability path to a closed class. By finiteness, there are an integer $r \geq 1$ and a real $\eta > 0$ such that every row sum of T_0^r is at most $1 - \eta$. Taking $0 < \eta < 1$ if necessary, the maximum-row-sum norm satisfies

$$\|T_0^m\|_\infty \leq (1 - \eta)^{\lfloor m/r \rfloor} \longrightarrow 0.$$

(The assertion is vacuous if there are no transient states.) But $P^2 = P$ implies $T_0^2 = T_0$, so $T_0 = 0$.

For a closed irreducible block C_a , each real vector x with $C_a x = x$ is constant: choose a maximal coordinate; its expression as a convex combination forces the same maximum at every successor, and irreducibility propagates it to all coordinates. Every column of C_a is fixed by C_a , since $C_a^2 = C_a$. Therefore all its columns are constant, and $C_a = \mathbf{1}\nu_a$ for a probability row vector ν_a . Its entries are positive by irreducibility. The equation $P^2 = P$ now gives $B = BC$, so the portion of any transient row on C_a is a nonnegative scalar multiple of ν_a .

Extend each ν_a by zero to a row of V , and set $U_{ia} = \sum_{j \in C_a} P_{ij}$. The preceding description proves $UV = P$. Both matrices have row sums one. For $i \in C_a$, row i of U is the a th coordinate row, so $VU = I$. These identities imply that the number of classes is $\text{rank } P = p$. $\square$

The rows of U at transient states need not be coordinate rows. Replacing this factorization by a partition quotient would discard real information and would make the next theorem false in its intended generality.

6.2 Reconstructing the effective generator

Theorem 6.2 (Effective-generator reconstruction). *Fix a critical α , with endpoints P, Q of ranks $p > q$, and take the canonical splitting $P = UV$ from Lemma 6.1. Set*

$$H(c) = VK_\alpha(c)U.$$

There is a unique real $p \times p$ row Laplacian G_α such that

$$H(c) = c(cI_p + G_\alpha)^{-1}, \quad K_\alpha(c) = U c(cI_p + G_\alpha)^{-1}V \quad (c > 0). \quad (6.1)$$

It is given explicitly by

$$G_\alpha = H(1)^{-1} - I_p. \quad (6.2)$$

Its zero eigenvalue is semisimple, its kernel has dimension q , and its zero-eigenvalue projection is $\hat{Q} = VQU$. In particular $\text{rank } G_\alpha = p - q$. Ordering the closed classes differently only permutes G_α .

Proof. Because $U, V, K_\alpha(c)$ are stochastic, so is $H(c)$. Equations (5.8) give $K_\alpha(c) = UH(c)V$, while (5.3) gives

$$H(c)H(e) = \frac{eH(c) - cH(e)}{e - c}. \quad (6.3)$$

Also $H(c) \rightarrow VPU = I_p$ as $c \rightarrow \infty$. Choose a sufficiently large real e so that $H(e)$ is invertible. Rearranging (6.3) yields

$$H(c)[cI_p + e(H(e)^{-1} - I_p)] = cI_p$$

for $c \neq e$. At $c = e$ the bracket is $eH(e)^{-1}$, so the same identity holds directly. Define $G_\alpha = e(H(e)^{-1} - I_p)$. A square matrix with a right inverse is invertible, so the first identity of (6.1) holds for every $c > 0$. It then gives (6.2) and uniqueness.

Since $H(c)\mathbf{1} = \mathbf{1}$, its inverse also fixes $\mathbf{1}$, and $G_\alpha\mathbf{1} = 0$. For large real c ,

$$H(c) = I_p - c^{-1}G_\alpha + O(c^{-2}).$$

Every off-diagonal entry of $H(c)$ is nonnegative. Multiplication by c and passage to the real limit therefore gives $(G_\alpha)_{ab} \leq 0$ for $a \neq b$. Thus it is a real Markov row Laplacian.

The limit as $c \downarrow 0$ is $\hat{Q} = VQU$. It is an idempotent, and $U\hat{Q}V = Q$, so its rank is q . From $G_\alpha H(c) = c(I - H(c))$ and boundedness of $H(c)$, we obtain $G_\alpha \hat{Q} = 0$. Conversely, if $G_\alpha x = 0$, then $H(c)x = x$ for all c and hence $\hat{Q}x = x$. Thus the kernel is exactly $\text{im } \hat{Q}$. The same argument on the other side gives $\hat{Q}G_\alpha = 0$.

There cannot be a nontrivial Jordan chain $G_\alpha y = x \neq 0$, $G_\alpha x = 0$. Such a chain would give $H(c)y = y - x/c$, which is unbounded as $c \downarrow 0$, contrary to the bounded matrix $H(c)$. A longer nontrivial Jordan chain contains one of length two. This proves semisimplicity and all remaining claims. $\square$

Corollary 6.3 (A real semigroup on each retract). *For real $u \geq 0$ define*

$$S_\alpha(u) = Ue^{-uG_\alpha}V.$$

These matrices are stochastic, $S_\alpha(0) = P$, $S_\alpha(u)S_\alpha(v) = S_\alpha(u+v)$, and $S_\alpha(u) \rightarrow Q$ as $u \rightarrow \infty$. Their ordinary real Laplace transform is

$$\int_0^\infty ce^{-cu}S_\alpha(u) du = K_\alpha(c).$$

Proof. Choose a real $M > \max_a (G_\alpha)_{aa}$. Then $T = I - G_\alpha/M$ is stochastic, and the ordinary absolutely convergent identity

$$e^{-uG_\alpha} = e^{-Mu} \sum_{r \geq 0} \frac{(Mu)^r}{r!} T^r$$

proves nonnegativity and row sums one. The semigroup identity follows from $VU = I$. To see the large- u limit, all nonzero eigenvalues of a real row Laplacian have strictly positive real part: if $Gz = \lambda z$ and $|z_a|$ is maximal and nonzero, write $q_{ab} = -G_{ab} \geq 0$ for $b \neq a$. Then

$$\text{Re } \lambda = \sum_{b \neq a} q_{ab}(1 - \text{Re}(z_b/z_a)) \geq 0,$$

because $|z_b/z_a| \leq 1$. Equality forces $z_b/z_a = 1$ for every positive outgoing rate. The original complex row equation then gives $\lambda = 0$, also when that row has no outgoing rate. The zero eigenvalue is semisimple by the preceding theorem, so the exponential converges to $\hat{Q}$. The Laplace transform identity follows by integrating the finite-dimensional real resolvent, or directly by differentiating $e^{-cu}e^{-uG_\alpha}$ and evaluating its endpoints. $\square$

Remark 6.4. This is a theorem about a *real* effective semigroup. It is not a claim that $S_\alpha(u)$ is the residue of a previously defined global surreal exponential $\exp(-ut^{-\alpha}L)$. No such global analytic construction is required.

7 The converse: realizing a prescribed hierarchy

This section is the principal inverse result. It first realizes arbitrary projection flags explicitly, then realizes prescribed effective generators at every transition. The extra difficulty in the second construction is entrywise positivity, not spectral decomposition.

7.1 Every stochastic projection flag occurs

Theorem 7.1 (Projection-flag realization). *Let $P_0 = I, P_1, \dots, P_m$ be distinct real stochastic idempotents satisfying*

$$P_i P_j = P_j P_i = P_{\max(i,j)}, \quad \text{rank } P_m = 1.$$

Choose any $\alpha_1 < \dots < \alpha_m$ in Γ , and write $\tau_j = t^{\alpha_j}$. Then

$$H = \sum_{j=1}^m \tau_j (P_{j-1} - P_j) \tag{7.1}$$

is a Hahn row Laplacian. Its normalized resolvent is exactly

$$R_H(s) = P_m + \sum_{j=1}^m \frac{s}{s + \tau_j} (P_{j-1} - P_j). \tag{7.2}$$

Its plateaux are precisely the prescribed P_j , and its crossover at α_j is

$$K_{\alpha_j}(c) = P_j + \frac{c}{c+1} (P_{j-1} - P_j). \tag{7.3}$$

An irreducible row Laplacian with the same shadows at every scale is obtained by the completion in Lemma 7.2 below.

Proof. Put $E_j = P_{j-1} - P_j$. These are algebraic idempotents with $E_i E_j = 0$ for $i \neq j$, $E_j P_m = P_m E_j = 0$, and $P_m + \sum_j E_j = I$. Each E_j has positive rank. They need not be nonnegative. The alternative expression

$$H = \tau_1 I + \sum_{j=1}^{m-1} (\tau_{j+1} - \tau_j) P_j - \tau_m P_m \tag{7.4}$$

shows that every off-diagonal entry is nonpositive, because $0 < \tau_{j+1} < \tau_j$. Also $H\mathbf{1} = 0$. Multiplying (7.2) by $sI + H$ proves the exact identity.

At scale $s = ct^{\alpha_j}$, the residue of $s/(s + \tau_i)$ is zero for $i < j$, $c/(c+1)$ for $i = j$, and one for $i > j$. Telescoping the last terms gives (7.3). At a scale strictly between successive α_j , the same calculation gives P_j . Since $E_j \neq 0$, no stated crossover disappears and there are no others. $\square$

Lemma 7.2 (Irreducible completion invisible to the hierarchy). *Suppose a Hahn row Laplacian H has final shadow $P_m = \mathbf{1}\pi_0$ at all scales strictly above α_m . Choose $\beta > \alpha_m$, let $\epsilon = t^\beta$, and let $J = \mathbf{1}\nu$ for any strictly positive real probability row ν . Then*

$$L = H + \epsilon(I - J) \quad (7.5)$$

has strictly negative off-diagonal entries. If H has the hierarchy of Theorem 7.1, or of Theorem 7.4 below, L has the same shadow at every α and every $c > 0$.

Proof. Strict negativity follows from the added rate $\epsilon\nu_j > 0$ on every edge. For every positive s , a direct rank-one inverse computation gives

$$R_L(s) = (sI + \epsilon J)((s + \epsilon)I + H)^{-1} \quad (7.6)$$

$$= \left(\frac{s}{s + \epsilon} I + \frac{\epsilon}{s + \epsilon} J \right) R_H(s + \epsilon). \quad (7.7)$$

For completeness, if $M = (s + \epsilon)I + H$, then $M^{-1}J = J/(s + \epsilon)$ because $H\mathbf{1} = 0$. Thus $(M - \epsilon J)^{-1} = (I + \epsilon J/s)M^{-1}$, proving (7.6). The order of these factors matters; no commutation of J and H is assumed.

If $s = ct^\alpha$ and $\alpha < \beta$, the first factor in (7.7) has residue I , and $s + \epsilon$ has the same leading term as s . The leading-forest argument, valid also for reducible H , shows that the shadow is unchanged. If $\alpha \geq \beta$, the parameter $s + \epsilon$ has valuation β , so its H -shadow is P_m . The first factor has residue a stochastic combination of I and J . Since $JP_m = P_m$, the product still has residue P_m . In particular the new edge scale β creates no new leading crossover. $\square$

7.2 Admissible effective data

Let a flag $P_0, \dots, P_m$ as above be given. For each j , choose the canonical splitting

$$P_{j-1} = U_j V_j, \quad V_j U_j = I_{p_{j-1}}, \quad p_j = \text{rank } P_j,$$

and set

$$Q_j = V_j P_j U_j. \quad (7.8)$$

It is a stochastic idempotent of rank p_j , and $U_j Q_j V_j = P_j$.

Definition 7.3 (Compatible effective generators). A family of real row Laplacians G_j of sizes p_{j-1} is compatible with the flag if

$$\lim_{c \downarrow 0} c(cI + G_j)^{-1} = Q_j. \quad (7.9)$$

Equivalently, Q_j is the zero-eigenvalue projection of G_j .

There is always at least one compatible choice, namely $G_j = I - Q_j$. But the next theorem allows any compatible G_j , including nonreversible generators, complex eigenvalues, unequal eigenvalues within a single scale, and nontrivial Jordan blocks at nonzero eigenvalues.

Theorem 7.4 (Prescribed-crossover realization). *Fix any flag $P_0 = I, P_1, \dots, P_m$ of distinct stochastic idempotents ending in rank one, any compatible real generators G_j , and any prescribed scales $\alpha_1 < \dots < \alpha_m$ in a nonzero divisible ordered group Γ . There exists an irreducible Hahn row*

Laplacian L whose only leading resolvent crossovers are those scales, whose plateaux are the P_j , and whose shadows at them are exactly

$$K_{\alpha_j}(c) = U_j c(cI_{P_{j-1}} + G_j)^{-1} V_j \quad (c > 0). \quad (7.10)$$

It can be chosen with finite-support rates. Moreover, after replacing every existing rate by its leading monomial, it can be chosen with positive monomial rates while preserving all these shadows.

Proof. Step 1: lift the effective generators. Set

$$E_j = P_{j-1} - P_j, \quad A_j = U_j G_j V_j.$$

Compatibility implies $G_j Q_j = Q_j G_j = 0$, and hence

$$A_j = E_j A_j E_j, \quad A_j \mathbf{1} = 0. \quad (7.11)$$

On the real subspace $\text{im } E_j$, the operator A_j is invertible. Indeed the restriction is identified through U_j, V_j with the restriction of G_j to $\ker Q_j$, where zero is not an eigenvalue. The mutually annihilating idempotents $E_1, \dots, E_m, P_m$ therefore decompose the whole vector space into the intended spectral blocks. In particular $A_i A_j = 0$ for $i \neq j$.

The naive sum $\sum_j t^{\alpha_j} A_j$ need not be a row Laplacian. A lifted A_j may have a positive off-diagonal entry on a pair of original states in the same preceding class. The next step repairs this without altering a prescribed crossover.

Step 2: insert positivity corrections between scales. For $2 \leq j \leq m$ choose $\alpha_{j-1} < \beta_j < \alpha_j$, possible by divisibility. Put

$$H = \sum_{j=1}^m t^{\alpha_j} A_j + \sum_{j=2}^m t^{\beta_j} (I - P_{j-1}). \quad (7.12)$$

Every row sum is zero. We check the off-diagonal signs carefully. If $(P_{j-1})_{ab} = 0$, nonnegativity of U_j, V_j implies $(U_j)_{ar}(V_j)_{rb} = 0$ for each r . Thus all diagonal contributions to $(U_j G_j V_j)_{ab}$ vanish, and its remaining contributions are nonpositive. Consequently

$$(A_j)_{ab} > 0 \implies (P_{j-1})_{ab} > 0 \quad (a \neq b). \quad (7.13)$$

For $j = 1$ the preceding projection is I , so $A_1 = G_1$ has no positive off-diagonal entries. For $j \geq 2$, any positive contribution $t^{\alpha_j}(A_j)_{ab}$ is preceded by the strictly negative contribution $-t^{\beta_j}(P_{j-1})_{ab}$ of smaller valuation. All exponents in the ordered list

$$\alpha_1 < \beta_2 < \alpha_2 < \dots < \beta_m < \alpha_m$$

are distinct. Therefore the earliest nonzero coefficient in any off-diagonal entry of H cannot be positive: every possible positive coefficient has an earlier negative one. The leading coefficient is negative, or the entry is zero. This proves that H is a row Laplacian. It is a finite sum of monomials.

Step 3: show that the inserted scales are invisible. On $\text{im } E_i$, the operator $I - P_{j-1}$ is the identity if $i < j$ and zero if $i \geq j$. Thus the block of H on $\text{im } E_i$ is

$$t^{\alpha_i} A_i + r_i I_{\text{im } E_i}, \quad r_i = \sum_{j>i} t^{\beta_j}. \quad (7.14)$$

Here $r_m = 0$, and otherwise $v(r_i) > \alpha_i$. The block on $\text{im } P_m$ is zero. Use any fixed real basis of each $\text{im } E_i$. The normalized resolvent on that block, at $s = ct^\alpha$, is

$$ct^\alpha(ct^\alpha I + t^{\alpha_i} A_i + r_i I)^{-1}.$$

If $\alpha < \alpha_i$, factor out ct^α ; the residue is the identity. If $\alpha > \alpha_i$, factor out t^{α_i} ; invertibility of the real A_i shows that the residue is zero. If $\alpha = \alpha_i$, the residue is $c(cI + A_i|_{\text{im } E_i})^{-1}$. This last inverse exists: under the retract isomorphism, $A_i|_{\text{im } E_i}$ is the restriction of G_i to its invariant nonzero block, and $cI + G_i$ is invertible for every $c > 0$ by the real matrix-forest identity, including its reducible case. These residue statements therefore use only inversion of a matrix with invertible real residue. The only crossovers are the α_i , and between them the shadows telescope to the prescribed P_i .

At α_j the later blocks together with P_m sum to P_j . Under U_j, V_j , this is the zero block Q_j of G_j , while $\text{im } E_j$ is its complement. Consequently the full shadow is exactly (7.10). Since $p_{j-1} > p_j$, that function has different endpoints and the crossover is nontrivial.

Step 4: impose irreducibility and then monomial rates. Apply Lemma 7.2 at a scale above α_m . The resulting L has a positive rate on every ordered pair and still has finite support. Each rate is its positive leading monomial times $1 + \varepsilon_e$ with $v(\varepsilon_e) > 0$. Theorem 4.1 already proves that replacing it by that leading monomial changes none of the shadows. The stronger relative statement will be established in Theorem 8.1. $\square$

Corollary 7.5 (A necessary and sufficient criterion). *A finite collection of plateau projections and real crossover kernels is the leading resolvent hierarchy of an irreducible finite surreal rate matrix if and only if its plateaux form a strict nested stochastic-idempotent flag from I to rank one, and each crossover, on the preceding stochastic retract, is the normalized resolvent of a real Markov generator whose final projection is the following plateau. Any prescribed strictly ordered Hahn locations for the crossovers can be used.*

Proof. Necessity is Theorems 5.2 and 6.2. Sufficiency is Theorem 7.4, transported into **No** by (2.1) whenever the chosen scales are surreal exponents. $\square$

7.3 The reversible boundary

Proposition 7.6 (Full-support reversible flags). *Suppose a surreal rate matrix is reversible with stationary probability π and every entry of $\pi_0 = \text{res } \pi$ is strictly positive. Then every plateau is the conditional-expectation matrix of a partition, weighted by π_0 , and the partitions are nested by coarsening. Conversely, fix a strictly positive real probability row π_0 and a finite strictly coarsening chain of partitions $\mathcal{P}_0, \dots, \mathcal{P}_m$, starting with the discrete partition and ending with the one-block partition. Their π_0 -weighted conditional-expectation matrices are precisely the plateaux of a reversible irreducible Hahn generator at any prescribed scales $\alpha_1 < \dots < \alpha_m$ in Γ .*

Proof. Detailed balance gives $\text{diag}(\pi)R_L(s) = R_L(s)^T \text{diag}(\pi)$. Taking residues shows that each plateau P is self-adjoint for the positive real weights π_0 . It also preserves π_0 . The splitting proof shows that any transient state of a stochastic idempotent has a zero column. Such a state cannot have positive mass under a stationary probability, so there are no transient states. Each closed block is $\mathbf{1}_{\nu_a}$, and detailed balance forces $\nu_a(j) = \pi_0(j)/\pi_0(C_a)$ on its class. This is precisely weighted conditional expectation onto that partition. The range consists of observables constant on each block; nested ranges give coarsening partitions.

Conversely, for each partition $\mathcal{P}_j$ define

$$(P_j)_{ab} = \begin{cases} \pi_0(b)/\pi_0(B), & a, b \in B \in \mathcal{P}_j, \\ 0, & \text{otherwise.} \end{cases}$$

Positive weights make these matrices well-defined stochastic idempotents, self-adjoint for π_0 . Coarsening gives $P_i P_j = P_j P_i = P_{\max(i,j)}$. The endpoints give $P_0 = I$ and $P_m = \mathbf{1}\pi_0$, and strict coarsening makes their ranks strictly decrease. Use (7.1) for these projections. Every summand is self-adjoint for π_0 and annihilates $\mathbf{1}$, so detailed balance holds and π_0 is stationary. Since P_m has strictly positive entries, (7.4) makes every off-diagonal entry strictly negative when $m \geq 1$. Theorem 7.1 supplies exactly the prescribed plateaux and crossovers. If the state space has one element, the chain has $m = 0$; the zero generator realizes its sole plateau, with no crossover. $\square$

The full-support condition cannot be removed. Example 11.1 gives an exactly reversible system whose limiting projection has a genuinely mixed transient row. The broader stochastic-retract theorem is not just a reformulation of the reversible partition case.

8 Uniform relative stability at every scale

Resolvent perturbation estimates based on a norm of $(sI + L)^{-1}$ can lose precision near a small spectral gap. Positivity allows a different statement: control relative edge errors by relative *entry* errors, simultaneously for all positive s .

Theorem 8.1 (All-parameter, entrywise relative stability). *Let L, L' have the same strongly connected allowed graph. Suppose*

$$q'_e = q_e(1 + \eta_e), \quad v(\eta_e) \geq \delta > 0 \quad (e \in E).$$

For every positive $s \in F$, and every i, j ,

$$v\left(\frac{R_{L'}(s)_{ij}}{R_L(s)_{ij}} - 1\right) \geq \delta. \quad (8.1)$$

More generally the same bound holds with $R_{L'}(s')$ in the numerator if $s' = s(1 + \eta_s)$ and $v(\eta_s) \geq \delta$. There is no dependence on a spectral gap, on s , or on the valuation of the entry being compared.

Proof. For every forest f ,

$$q'(f)(s')^{d-|f|} = q(f)s^{d-|f|} \prod_{e \in f} (1 + \eta_e)(1 + \eta_s)^{d-|f|} = q(f)s^{d-|f|}(1 + O_v(\delta)).$$

Let Z be either the denominator forest sum $D(s)$ or any marked numerator $N(s)_{ij}$, and let Z' be its perturbed counterpart. Every summand of Z is positive. If their minimum valuation is ζ , then $v(Z) = \zeta$ without cancellation, whereas $v(Z' - Z) \geq \zeta + \delta$. Hence $Z'/Z = 1 + O_v(\delta)$. Both numerator and denominator are nonzero, because the graph is strongly connected. Dividing the two relative identities gives (8.1); the inverse of $1 + O_v(\delta)$ is again $1 + O_v(\delta)$. $\square$

Corollary 8.2 (Leading data suffice simultaneously). *Two positive rate systems with the same graph, edge valuations, and edge leading coefficients have identical plateau projections, crossover kernels, effective generators, and leading terms of all marked entries at every scale. The assertion holds simultaneously over all positive resolvent parameters, not only at the finitely many critical scales.*

Proof. For finitely many edges, the relative discrepancies have a common positive valuation lower bound. Apply Theorem 8.1. The effective generators are determined by the compressed kernels in (6.2). $\square$

Example 8.3 (Sharpness of the valuation bound). Take two states, $q_{12} = q_{21} = 1$, and $s = 1$. Then $R_L(1)_{12} = 1/3$. Change only q_{12} to $1 + \eta$, with $v(\eta) = \delta > 0$. The new entry is $(1 + \eta)/(3 + \eta)$, and its relative error is exactly

$$\frac{(1 + \eta)/(3 + \eta)}{1/3} - 1 = \frac{2\eta}{3 + \eta}.$$

Its valuation is δ . Thus the exponent in (8.1) cannot be increased in general.

Remark 8.4 (Why the hypotheses matter). A new edge is not a relative perturbation of a zero edge. It may change the hierarchy, unless an additional argument such as Lemma 7.2 shows otherwise. Likewise, positive leading forest sums are essential: for a signed sum, near cancellation can amplify relative errors. The theorem is not an arbitrary-matrix resolvent condition-number bound.

9 A complete finite diagram has a rank-one realization

Higher rank is genuine, but the finite nature of a forest calculation creates a precise limitation on what it can detect. This section preserves not merely the critical eigenvalue scales, but every marked entry's leading forest choices.

9.1 Finite sign separation

Lemma 9.1 (Finite positive signs can be represented rationally). *Let W be a finite-dimensional rational vector space with an ordered-abelian-group order, and let $x_1, \dots, x_r$ be a finite family of strictly positive elements. There exists a $\mathbb{Q}$ -linear map $\ell : W \rightarrow \mathbb{Q}$ such that $\ell(x_i) > 0$ for every i . No assertion of positivity on all of $W_{>0}$ is made.*

Proof. If the list is empty, take the zero functional. Otherwise choose a rational basis of W and regard the x_i as rational vectors in $\mathbb{R}^a$. Their convex hull cannot contain zero. Otherwise a convex representation of zero can be chosen with a minimal number of nonzero coefficients. The augmented vectors $(x_i, 1)$ in that representation are linearly independent: a dependence would allow the coefficients to be varied while preserving their sum and the represented vector, until one coefficient became zero. Solving the resulting independent rational linear system makes the minimal representation rational. It would therefore express zero as a nonempty positive rational combination of positive elements of the ordered group, a contradiction.

The convex hull is compact and excludes zero. If y is its point of minimal Euclidean norm, the usual nearest-point inequality gives $y \cdot x_i \geq \|y\|^2 > 0$ for each i . A sufficiently close rational vector has the same finitely many strict inequalities. Dot product with it defines the required rational functional. $\square$

This elementary finite separation principle is used as an input, not claimed as new. Higher-rank Hahn specialization in more general tropical settings is studied in [9].

9.2 The marked-forest arrangement

For every forest f define an affine function on the ordered group,

$$\lambda_f(\alpha) = w(f) + (d - |f|)\alpha.$$

All coefficient data belong to the finite-dimensional space

$$W = \text{span}_{\mathbb{Q}}\{\gamma_e : e \in E\} \subseteq \Gamma,$$

a rational subspace of Γ because Γ is divisible (item (D3) of Section 2.3). If two forest functions have different slopes, their intersection is a point of W . Let T be the finite set of all such intersection points. We retain intersections that are not active for the denominator, since some of them may matter to a marked numerator. Forest functions with the same slope are ordered by their constant terms, or coincide identically.

Theorem 9.2 (Simultaneous rank-one realization of the leading diagram). *For any fixed finite strongly connected Hahn rate system there is a $\mathbb{Q}$ -linear map $\ell : W \rightarrow \mathbb{Q}$ such that the ordinary positive rates*

$$q_e(x) = a_e x^{\ell(\gamma_e)}, \quad x > 0, \quad (9.1)$$

reproduce its entire finite marked-forest leading diagram as $x \downarrow 0$. Specifically, the ordered cells and crossing points of the forest arrangement correspond, and at corresponding locations they have the same minimizing forests for the denominator and for every marked numerator. Consequently they have the same rational amplitude formulas, plateau matrices, crossover matrices, and effective generators.

At a crossing $\alpha \in T$, its numerical location is $\ell(\alpha)$. On an open cell, the corresponding affine exponent formulas are obtained by applying ℓ to their constant terms and replacing the scale variable by a real variable in the corresponding cell. This is not a global order embedding of Γ . After a change $x = y^M$, all exponents in the rates can be chosen integral, possibly negative.

Proof. Form a finite list of positive elements of W consisting of all positive differences between distinct points of T , and all positive differences $w(f) - w(g)$ between equal-slope forest functions. Optional finitely many positive edge-valuation differences or other prescribed comparisons may be added. Apply Lemma 9.1. It preserves the strict order of the crossing points and the signs of equal-slope differences. Equalities are automatically preserved by linearity.

For two different-slope functions, their difference is an integer multiple of $\alpha - \theta$, where $\theta \in T$ is their intersection. Thus its sign on any cell, or at a crossing, is determined by the ordering relative to θ . That ordering is preserved. Together with the equal-slope comparisons, this shows that the total comparison pattern of all forest functions is preserved in each cell and at each crossing. Restricting the family to any set $r_f(i) = j$ preserves its minimizers as well.

For the rates (9.1), a forest term at the numerical observation scale $s = cx^a$ has exponent $\ell(w(f)) + (d - |f|)a$ and coefficient $A(f)c^{d-|f|}$. Finite sums of positive monomials are asymptotic to the sum of the terms with smallest exponent. Since the minimizing forests and their coefficients agree, all amplitude formulas in Theorem 4.1 agree. In particular their matrix residues, plateau splittings, and the reconstructed real generators agree.

There are finitely many rational edge exponents. A common positive denominator M makes the exponents of $q_e(y^M)$ integral. If desired, multiplying all rates by the same power of y makes them nonnegative, at the cost of translating every observation-scale exponent by that same amount. $\square$

Remark 9.3 (Precisely what is and is not preserved). An arbitrary $\alpha \in \Gamma$ need not belong to W , so an expression $\ell(\alpha)$ is not used for every possible scale. The finite arrangement provides the correspondence: select a real scale in the matching open cell, or use $\ell(\alpha)$ at a crossing. The marked exponent $m_{ij,\alpha} - m_\alpha$ is a piecewise affine expression. Its combinatorial formula and amplitude are preserved; its value as an element of Γ is not literally identified with a real number unless the relevant arguments are in the domain of ℓ . The theorem preserves leading data, not complete Hahn series, all possible inequalities, or the entire ordered group.

Example 9.4 (Why the finite qualification is sharp). In $\Gamma = \mathbb{Q}^2$ with lexicographic order put $\delta = (0, 1)$ and $\beta = (1, 0)$. Then

$$0 < N\delta < \beta \quad \text{for every positive integer } N.$$

No $\mathbb{Q}$ -linear map $\Gamma \rightarrow \mathbb{R}$ can preserve all these inequalities while sending δ to a positive number: it would require the finite real number $\ell(\beta)$ to exceed $N\ell(\delta)$ for every N . A finite family of inequalities can be preserved, for example by sending δ to 1 and β to a sufficiently large integer. Theorem 9.2 uses exactly such a finite list, not its infinite continuation.

10 The surcomplex spectrum and the forest convexity law

Because $K = \mathbb{C}((t^\Gamma))$ is algebraically closed, which uses the divisibility of Γ , the eigenvalues of L lie in a surcomplex Hahn workspace. The tree coefficient $\sigma_d > 0$ in (3.3) shows that zero has algebraic multiplicity one. Denote the other eigenvalues, with algebraic multiplicity, by $\lambda_1, \dots, \lambda_d$, and put $\zeta_r = v(\lambda_r)$.

10.1 Initial polynomials without a real Newton polygon

Lemma 10.1 (Valuations and initial eigenvalue factors). *For every $\alpha \in \Gamma$,*

$$m_\alpha = \sum_{r=1}^d \min(\alpha, \zeta_r). \quad (10.1)$$

If $\rho_r = \text{res}(t^{-\alpha}\lambda_r)$ for the indices with $\zeta_r = \alpha$, then

$$D_\alpha(c) = C_\alpha c^{\#\{r: \zeta_r > \alpha\}} \prod_{\zeta_r = \alpha} (c + \rho_r), \quad (10.2)$$

where $C_\alpha \neq 0$ is the product of the leading coefficients of the faster factors. Consequently the critical scales are exactly the distinct valuations of nonzero eigenvalues, including valuation zero when it occurs.

Proof. As a polynomial over K , $D(s) = \prod_r (s + \lambda_r)$. Replace s by $t^\alpha z$, where z is an indeterminate. The minimum coefficient valuation of a product of nonzero polynomials is the sum of those minima: after normalizing each factor to have minimum zero, the product of their nonzero residue polynomials remains nonzero in the integral domain $\mathbb{C}[z]$.

The normalized residue of $t^\alpha z + \lambda_r$ is respectively a nonzero constant, $z + \rho_r$, or z , according as $\zeta_r < \alpha$, $\zeta_r = \alpha$, or $\zeta_r > \alpha$. This proves both formulas. If no ζ_r equals α , the initial polynomial is a monomial. If at least one does, its largest and smallest nonzero powers differ, since all those ρ_r are nonzero. Hence it is not a monomial. By (4.3), being nonmonomial is exactly criticality. $\square$

Remark 10.5 below compares this lemma with Newton's root-valuation rule in the collection's polynomial-algebra report.

10.2 A characteristic polynomial for each effective chain

Theorem 10.2 (Effective characteristic and amplitude identity). *At a critical scale α , let k_-, k_+, p, q be as in Theorem 5.2. The effective generator satisfies the exact real polynomial identity*

$$\det(cI_p + G_\alpha) = \frac{cD_\alpha(c)}{b_{k_-}}. \quad (10.3)$$

Exactly $p - q$ nonzero eigenvalues of L have valuation α , counting algebraic multiplicity. Their normalized leading coefficients are exactly the nonzero eigenvalues of G_α , again counting multiplicity. In particular these coefficients have strictly positive real parts, and every nonzero surcomplex eigenvalue satisfies

$$v(\operatorname{Re} \lambda) = v(\lambda). \quad (10.4)$$

Proof. The trace relation in (3.10) gives

$$\operatorname{tr} K_\alpha(c) = \frac{\sum_{k \in I_\alpha} (n - k) b_k c^{d-k}}{D_\alpha(c)} = 1 + c \frac{D'_\alpha(c)}{D_\alpha(c)}. \quad (10.5)$$

On the other hand, cyclicity of trace and $VU = I$ give

$$\operatorname{tr} K_\alpha(c) = \operatorname{tr}(c(cI_p + G_\alpha)^{-1}) = c \frac{d}{dc} \log \det(cI_p + G_\alpha).$$

Therefore the logarithmic derivatives of $\det(cI_p + G_\alpha)$ and $cD_\alpha(c)$ agree for positive c . Their ratio is a constant rational function. The former is monic of degree p ; the latter has degree $1 + d - k_- = p$ and leading coefficient b_{k_-} . This proves (10.3) as a polynomial identity.

The smallest power of D_α is $d - k_+ = q - 1$, and the largest is $p - 1$. Comparing with (10.2), there are $q - 1$ slower nonzero eigenvalues and $p - q$ at the current scale. Cancelling the q zero factors in (10.3) identifies the other eigenvalues with the ρ_r of that factorization. Nonzero eigenvalues of a real row Laplacian have strictly positive real parts, as proved in Corollary 6.3. Thus the leading real part of every normalized λ_r is nonzero and positive, proving (10.4). $\square$

There is no assertion here that a nonreversible spectrum is real. The leading amplitudes can occur in nonreal conjugate pairs, as in Example 11.2. Nor is diagonalizability at nonzero eigenvalues needed for the determinant argument.

Theorem 10.3 (Discrete convexity and all eigenvalue valuations). *The forest profile satisfies*

$$h_1 - h_0 \leq h_2 - h_1 \leq \cdots \leq h_d - h_{d-1}. \quad (10.6)$$

These d increments are precisely the valuations of nonzero eigenvalues of L in nondecreasing order, including repetitions. At any critical scale, I_α consists of every integer from k_- through k_+ ; no intermediate forest size is missing from the initial polynomial.

Proof. Apply the classical forest determinant formula to the real effective row Laplacian G_α . Its zero eigenvalue has multiplicity q , so the largest number of edges in one of its in-forests is $p - q$. Every smaller edge count occurs by deleting edges of such a forest. All associated coefficients of

$\det(cI_p + G_\alpha)$ are positive. Thus this polynomial has strictly positive coefficients at all degrees from q to p , inclusive. Equation (10.3) implies that D_α has a positive coefficient at every degree from $q - 1$ to $p - 1$. Its definition forces every integer from k_- to k_+ to belong to I_α .

As the scale increases from the initial to the final plateau, the active indices run from 0 to d . Any jump from k_- to k_+ therefore has all intermediate points on the same supporting line, and

$$h_k - h_{k-1} = \alpha \quad (k_- < k \leq k_+).$$

Successive jumps occur at strictly increasing scales. This proves (10.6). The multiplicity statement in Theorem 10.2 then identifies the increments with the eigenvalue valuation list. $\square$

Remark 10.4 (Relation to optimal-forest spectral formulas). In a rank-one small-noise model, forest costs are sums of edge exponents, and successive optimal costs give familiar eigenvalue exponents in the Freidlin–Wentzell tradition; see the discussion in [4]. The present proof works in an arbitrary ordered group and simultaneously identifies the full real effective characteristic polynomial. It does not claim that the underlying rank-one optimal-forest idea is new.

Remark 10.5 (Newton’s rule in the polynomial-algebra report). The expression m_α is the Newton profile at center zero of the monic polynomial $D(s)$, whose roots are the $-\lambda_r$. So Lemma 10.1 is also the case $P = D$ of Newton’s root-valuation rule (`polynomial:thm:newton`) and of the initial-polynomial root count with its profile formula (`polynomial:thm:initialroots`, `polynomial:eq:profileproduct`) in the collection’s polynomial-algebra report [11]. That report works in the same kind of workspace, with Γ set-sized and divisible. The direct proof of Lemma 10.1 given above is kept.

Theorem 10.3 says more than Newton’s rule gives for a general polynomial. There, root valuations are read from the lower boundary of the coefficient points, and some points may lie strictly above that boundary. For the forest polynomial of a positive rate matrix this does not happen: every point (k, h_k) lies on the lower boundary, and every index from k_- to k_+ is active at a critical scale. The input that rules out the general behaviour is the positivity of the forest coefficients of the real effective generator.

11 Worked examples

11.1 A nonreversible three-state system

Let $\delta > 0$, write $\tau = t^\delta$, and consider

$$L = \begin{pmatrix} 2 & -2 & 0 \\ -1 & 1 + 3\tau & -3\tau \\ -\tau & 0 & \tau \end{pmatrix}. \quad (11.1)$$

The graph is strongly connected, but is not reversible: there is an edge $2 \rightarrow 3$ without its reverse. Exact expansion gives

$$D(s) = s^2 + (3 + 4\tau)s + 9\tau + 3\tau^2.$$

The forest profile is

$$(h_0, h_1, h_2) = (0, 0, \delta), \quad (b_0, b_1, b_2) = (1, 3, 9).$$

Thus the nonzero eigenvalue scales are 0 and δ . Between them and after them, respectively, the plateau projections are

$$P = \begin{pmatrix} 1/3 & 2/3 & 0 \\ 1/3 & 2/3 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad Q = \mathbf{1} \begin{pmatrix} 1/9 & 2/9 & 2/3 \end{pmatrix}. \quad (11.2)$$

At the slower scale,

$$K_\delta(c) = \frac{cP + 3Q}{c + 3}. \quad (11.3)$$

Use

$$U = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1/3 & 2/3 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Compression reconstructs

$$G_\delta = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}.$$

Its nonzero eigenvalue is 3, so the slow eigenvalue of L has leading term $3t^\delta$. The fast eigenvalue has leading term 3. The characteristic identity reads $\det(cI_2 + G_\delta) = c(c + 3) = cD_\delta(c)/3$, because $D_\delta(c) = 3c + 9$.

One rare entry illustrates the marked formula. For $0 < \alpha < \delta$, the plateau entry P_{13} is zero. The leading numerator term for R_{13} is 6τ , whereas the denominator has leading term $3ct^\alpha$. Therefore

$$R_L(ct^\alpha)_{13} = \frac{2}{c} t^{\delta-\alpha} (1 + \text{an element of positive valuation}).$$

The vanishing shadow does not mean the transition is absent; it has a definite smaller leading scale and amplitude.

11.2 A rank-two hierarchy with a tied fast scale

In lexicographically ordered $\mathbb{Q}^2$, let $\delta = (0, 1)$, $\beta = (1, 0)$, $\tau = t^\delta$, and $\zeta = t^\beta$. Then $\zeta \ll \tau^N$ for every positive integer N . Let P average each pair $\{1, 2\}$ and $\{3, 4\}$, and let $J = \mathbf{1}\mathbf{1}^T/4$. The inverse construction gives

$$H = \tau(I - P) + \zeta(P - J).$$

The rate within a pair is $\tau/2 - \zeta/4 > 0$, and each cross-pair rate is $\zeta/4$. This is a reversible complete graph. Its nonzero eigenvalues are τ, τ, ζ , and

$$D(s) = (s + \tau)^2(s + \zeta).$$

Consequently

$$(h_0, h_1, h_2, h_3) = (0, \delta, 2\delta, 2\delta + \beta), \quad (b_0, b_1, b_2, b_3) = (1, 2, 1, 1).$$

The two crossovers are

$$K_\delta(c) = P + \frac{c}{c+1}(I - P), \quad K_\beta(c) = J + \frac{c}{c+1}(P - J).$$

The fast transition loses two dimensions at once. Tied scales cause no ambiguity and require no arbitrary tie breaking. A numerical monomial specialization can reproduce this entire finite diagram, but cannot preserve $\zeta \ll \tau^N$ for every N ; this is the distinction in Theorem 9.2.

11.3 Reversibility does not eliminate transient mixtures

Example 11.1 (A fuzzy plateau in a reversible system). Let $p, q > 0$ be real with $p + q = 1$, and let $\tau = t^\delta$, $\delta > 0$. Put

$$L = \begin{pmatrix} \tau & 0 & -\tau \\ 0 & \tau & -\tau \\ -p & -q & 1 \end{pmatrix}.$$

It is reversible for the exact stationary law

$$\pi = \frac{1}{1 + \tau}(p, q, \tau),$$

since the two edgewise detailed-balance identities are immediate. Its determinant factor is $D(s) = (s + \tau)(s + 1 + \tau)$. At scales $0 < \alpha < \delta$, its plateau is

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ p & q & 0 \end{pmatrix}.$$

The third row is a genuine probability mixture between the two surviving states. The canonical retract uses the first two coordinate rows for V and the first two columns of P for U . Its slow effective generator is

$$G_\delta = \begin{pmatrix} q & -q \\ -p & p \end{pmatrix}.$$

The residue stationary law $(p, q, 0)$ is not strictly positive. Thus this example satisfies exact reversibility while lying outside the full-support conclusion of Proposition 7.6.

11.4 Prescribed nonreal dynamics and an invisible positivity correction

Example 11.2 (Why the general inverse theorem needs its correction). For clarity take a rank-one parameter t with $v(t) > 0$, measuring exponents in units of $v(t)$. Prescribe a fast generator that cycles through the first three states and leaves the fourth fixed:

$$A_1 = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Its zero-eigenvalue projection is

$$P = UV, \quad U = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1/3 & 1/3 & 1/3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Prescribe the slow effective generator

$$G_2 = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}, \quad A_2 = UG_2V,$$

and place it at scale 2. The naive sum $A_1 + t^2 A_2$ is not a row Laplacian: its $(1, 3)$ entry is $2t^2/3 > 0$. The intermediate correction gives

$$H = A_1 + t(I - P) + t^2 A_2. \quad (11.4)$$

Now the offending entry is $-t/3 + 2t^2/3 < 0$, and all the other off-diagonal signs are correct by the proof of Theorem 7.4.

The exact nonzero characteristic factor is

$$D(s) = (s + 3t^2)((s + t)^2 + 3(s + t) + 3). \quad (11.5)$$

Thus the eigenvalues are

$$0, \quad 3t^2, \quad t + \frac{3}{2} + \frac{\sqrt{3}}{2}i, \quad t + \frac{3}{2} - \frac{\sqrt{3}}{2}i.$$

This supplies a genuine surcomplex pair at the fast scale. The complete resolvent can be written

$$R_H(s) = s((s + t)I + A_1)^{-1}(I - P) + U s(sI_2 + t^2 G_2)^{-1} V. \quad (11.6)$$

At scale 0 its shadow is $c(cI_4 + A_1)^{-1}$; at scale 2 it is $Uc(cI_2 + G_2)^{-1}V$. At the inserted scale 1, for every real $c > 0$, the shadow is just P . A new leading *edge* scale has repaired positivity without becoming a new leading *resolvent* scale.

The final projection is $\mathbf{1}(1/9, 1/9, 1/9, 2/3)$, and this particular corrected matrix is already irreducible. Multiplication of (11.6) by $sI + H$ provides a direct exact check independent of the asymptotic argument.

12 Computation, verification, and limitations

12.1 A finite exact procedure

The theory gives a direct algorithm once the leading edge data are available. Enumerate in-forests by assigning to each vertex either no outgoing edge or one allowed successor, and reject assignments containing a directed cycle. For each surviving forest, compute its root map, valuation cost, and positive amplitude. Grouping by edge count gives h_k, b_k, B_k ; grouping additionally by the mark $r_f(i) = j$ gives the rare-entry data.

The increments $h_k - h_{k-1}$ already give the critical scales with multiplicity, by Theorem 10.3. At a critical scale form $N_\alpha(c)/D_\alpha(c)$ and the preceding projection P . Find its closed classes to construct U, V , then use

$$G_\alpha = (VK_\alpha(1)U)^{-1} - I.$$

Thus a single real matrix inverse per scale reconstructs the effective generator once the positive forest coefficients are known. Computing surcomplex eigenvectors is unnecessary.

This enumeration is finite, not an efficiency claim. There are at most n^n parent assignments in a complete directed graph, and the number of forests grows rapidly. The exact comparison contract is also significant: one must be able to add and compare the input valuations, perform rational division in their span, and manipulate the real leading coefficients exactly. Arbitrary real coefficient names need not supply decidable equality; arbitrary surreal names need not expose the leading exponent. A mathematical finiteness theorem is not a uniform algorithm on every possible representation of a surreal number.

12.2 Reproducible checks included in this directory

The accompanying `code/verify.py` uses Python’s exact rational numbers and SymPy, with no floating-point tolerances and no external data. The recorded run used seed 20260922. It tested 48 complete directed graphs on two through five vertices, with positive rational leading coefficients and valuations in lexicographically ordered $\mathbb{Q}^2$. Across these instances it enumerated 17,280 forests and checked 105 critical kernels.

The checks include convexity of the optimal forest profile; equality of the set of its distinct increments with the set of critical scales; stochasticity, idempotence and nesting of all plateaux; the canonical stochastic retract identities; the signs and ranks of the reconstructed effective generators; and equality with their resolvents at the exact rational parameters $c = 1, 2, 3$. These checks use forests with tied costs as well as distinct costs. The program does not compare multiplicities of increments with multiplicities of eigenvalue valuations directly. It checks the rank of every plateau and of every reconstructed generator, and the rank drop at a scale is the multiplicity asserted by Theorem 10.2.

Separate symbolic checks verify the three-state determinant and crossover, the exact projection-flag inverse, the rank-one irreducible-completion identity, a four-state matrix-forest identity, the sharp relative-error example, and the prescribed-crossover construction in Example 11.2, including the absence of an extra crossover at its positivity-correction scale. The resulting `data/verification.json` records **PASS** and the individual instance sizes. These checks do not establish the general theorems: in particular they do not replace the arbitrary-rank separation proof or the positivity argument for all compatible effective generators. They are independent of the algebraic Lean coverage described in the opening status paragraph.

The program writes its output over `data/verification.json` next to it, so it should be run on a copy of this directory. Such a rerun at placement, under Python 3.14.4 and SymPy 1.14.0, reproduced the recorded file exactly except for operating-system line endings. The manuscript’s own build record, `data/build_validation.json`, names Python 3.13.5 and SymPy 1.14.0.

12.3 What has actually been settled here

The forward and inverse theorems settle an internally stated classification problem: the necessary and sufficient conditions for a *finite leading resolvent hierarchy* to arise from positive surreal rates. They include the realizability of arbitrary compatible reduced dynamics, rather than merely an ordered eigenvalue list. This is a definite theorem with a constructive proof, not a suggestion to investigate the problem later.

The marked formula and the relative bound give two different levels of information. The residue matrix describes effective stochastic motion. The marked entry’s valuation and amplitude describe transitions too rare to appear in that shadow. Relative stability controls both uniformly over the observation parameter. The characteristic identity, in turn, explains why the same forest data describe the leading surcomplex relaxation amplitudes.

The specialization theorem gives a complementary negative conclusion: this finite information does not distinguish higher rank from a suitably chosen rank-one model. Higher rank remains meaningful for an unbounded family of scale comparisons, for complete series rather than leading terms, and for data whose size is not fixed in advance. The finite theorem must not be extrapolated to those settings.

12.4 Remaining directions, not results claimed in this paper

The normalized resolvent does not determine a theory of surreal-valued sample paths, stochastic integration, or a globally defined surreal heat semigroup. Constructing such objects requires additional choices of convergence and summability structures. Likewise, no result here covers countably infinite state spaces, time-inhomogeneous rates, uniform estimates as $n \rightarrow \infty$, or a full hierarchy of large-deviation transition events. The collection's report on measures and probability (Section 1.5) is not used to supply any of these objects.

There are natural further questions. One is to develop efficient exact algorithms for the marked-forest diagram without exhaustive enumeration. Another is to identify conditions under which the real effective semigroups arise from a specified, independently defined Hahn-summable evolution. A third is to ask what additional finite data, beyond leading resolvent terms, must be retained to obstruct some rank-one specializations. These questions are separated from the proved realization theorem and are not presented as solved.

13 Conclusion

Positive surreal transition rates admit a particularly rigid finite theory once the observable is fixed. Their all-scale resolvent shadows are exactly finite compatible cascades of real Markov dynamics on nested stochastic retracts. The inverse construction shows that no unmentioned reversibility or partition condition is hidden in this description. Its positivity corrections can occupy new edge scales without creating new leading dynamical scales.

The same finite forest algebra supplies uniform relative perturbation control, rare-entry asymptotics, the effective characteristic polynomial, and an exact rank-one realization of the whole finite leading diagram. This is simultaneously a constructive theory of multiscale surreal Markov matrices and a precise limit on what finite leading-order observations can reveal about the rank of the surreal scale group.

A Proof dependencies and conventions at a glance

The only imported structural results about generalized numbers are Conway normal form, the standard Hahn-field arithmetic, and real/algebraic closedness for the stated divisible groups. The only imported combinatorial matrix identity is the directed matrix-forest theorem. The stochastic-idempotent structure theorem and the finite sign-separation lemma are reproved. All realization and stability arguments are then finite algebra.

Conclusion	Main dependencies
Leading and marked shadows	Positive finite forest sums; Proposition 3.1.
Nested stochastic flag	Resolvent identity; trace of a forest matrix.
Effective real generator	Stochastic retract; compressed resolvent identity.
Prescribed-crossover realization	Block support of compatible generators; sign-cover implication (7.13); intermediate corrections.
Irreducible completion	Exact rank-one inverse identity; rank-one final projection.
Relative stability	No cancellation in positive forest sums.
Rank-one specialization	Finite affine forest arrangement; finite rational sign separation.
Surcomplex amplitudes	Polynomial factorization over K ; effective characteristic identity.
Forest convexity	Positivity of every attainable forest coefficient of the real effective generator.

There is no circular reliance between the inverse and stability theorems. The inverse theorem's monomial replacement assertion already follows from the leading-forest formula. The later stability theorem strengthens it but is not needed to construct the hierarchy. Similarly, convexity is proved after the effective generator exists; convexity is convenient for computation but was not assumed in constructing that generator.

The time parameter is always distinguished from the scale exponent: $s = ct^\alpha$ is a Hahn killing rate, c is an ordinary positive real amplitude, and u in the effective semigroup is an ordinary nonnegative real time. The valuation is additive, so larger α means smaller s . Eigenvalues refer to the row Laplacian L , not to the conventional generator $-L$; nonzero real parts therefore have the positive sign.

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